

CONFIDENTIAL TECHNICAL & COST PROPOSAL

Delivery API Integration Strategy, Vendor Analysis & Cost Proposal

A comprehensive evaluation of automated courier integrations for Intrust India's e-commerce platform — covering pan-India logistics, Bhopal hyperlocal same-day delivery for grocery products, technical architecture, and transparent cost models for client review.

PREPARED FOR	PREPARED BY	TARGET GEOGRAPHY	VERSION & DATE
Intrust India	Intrust Tech Team	Bhopal & Pan-India	v1.0 • Sept 2026

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SECTION 01

Executive Summary

Intrust India currently manages order dispatch manually: merchants self-fulfill and manually copy-paste tracking numbers into the admin panel. To support multi-vendor expansion and enable **same-day local grocery delivery** across Bhopal alongside **pan-India shipping** for fashion/electronics, this proposal outlines an automated API-driven logistics infrastructure.

RECOMMENDED APPROACH 2-Provider Stack Shiprocket (National) + Shadowfax (Bhopal)	TOTAL DEV INVESTMENT ₹80K – 1.2 Lakh 18–21 days across 2 phased milestones	SHIPPING COST RECOVERY 100% Recoverable Standard checkout delivery fees offset couriers
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Architecture Feasibility: Intrust's Supabase database schema already contains the necessary tracking fields, delivery status enums, and update RPCs. Integration is purely additive with zero core database refactoring required.

SECTION 02

Current Architecture Assessment

The platform utilizes Next.js 14 App Router and a Supabase PostgreSQL backend on a dedicated VPS. The existing database structure is already primed to receive courier API integration:



EXISTING ASSET	DATABASE LOCATION / API ROUTE	DELIVERY API MAPPING
Delivery Status Enum	shopping_order_groups.delivery_status	Maps directly to partner tracking states
Tracking Identifier	shopping_order_groups.tracking_number	Auto-stores courier AWB / Waybill ID
Estimated Delivery	shopping_order_groups.estimated_delivery_at	Auto-populated via courier SLA response
Status Update RPC	update_order_delivery_v3()	Invoked by Webhook with idempotency
Customer WhatsApp Alerts	/api/orders/[orderId]/status/route.js	Fires automatically on shipped/delivered milestones
Product Routing Tag	products.delivery_type (standard vs. hyperlocal)	Additive column to route grocery vs. standard carts

SECTION 03

Delivery Models: Two Distinct Operating Strategies

Intrust operates a multi-vendor marketplace where different product types require distinct logistics patterns:

Model A: Merchant Direct Pickup

Recommended

On-Demand Pickup from Merchant's Local Store / Outlet

✓ Zero warehouse capital or physical lease expenditure required.

✓ Courier rider dispatched directly to merchant store upon "Pack & Ship".

✓ Delivers groceries in 2–4 hours locally, and standard goods in 1–3 days.

✓ Seamlessly fits current multi-vendor merchant architecture.

Fulfilment SLA:

2 – 24 Hours

Model B: Centralized Dark Store

Future Phase

Quick-Commerce Central Micro-Warehouse (Blinkit / Zepto Style)

✗ Requires dedicated warehouse lease, racking, and cold storage in Bhopal.

✗ Requires stock ownership, upfront working capital, and expiry management.

✗ High initial Capex barrier (₹15L – 30L minimum setup cost).

✓ Enables ultra-fast 15–30 minute delivery once scale exceeds 1,000 orders/day.

Recommended For:

Scale Stage (Phase 3+)

SECTION 04

Standard E-Commerce Logistics (Pan-India)

For non-perishable goods (fashion, electronics, dry goods). Couriers pick up from Bhopal merchants and deliver nationwide.

Shiprocket (Aggregator)

★ Recommended

Single API covering 25+ couriers (Delhivery, BlueDart, DTDC, Ekart)

✓ Single master contract & unified wallet for all national carriers.

✓ Smart AI recommendation picks best rate/speed courier per pincode.

✓ Automated Non-Delivery Report (NDR) engine reduces costly RTO losses.

✓ Robust developer sandbox, webhooks, and automated label generation.

Platform Fee: ₹499 – 799 / mo Rate: ₹40 – 110 / 500g

Delhivery (Direct Carrier)

Enterprise

Direct contract with India's largest integrated logistics player

✓ Extensive tier-2/3 network reach and dedicated fleet coverage.

✓ Volume discounts of 20–35% once exceeding 3,000+ orders/month.

✗ Direct B2B contract requires volume commitments and enterprise negotiation.

✗ Single carrier risk; no automated failover if local hubs experience backlogs.

Platform Fee: Custom Contract Rate: Negotiated SLA

SECTION 05

Hyperlocal Logistics: Bhopal Grocery & Same-Day

Grocery and perishable goods require dedicated 2-wheeler fleets with sub-4-hour SLAs across Bhopal (MP Nagar, Arera Colony, Kolar Road, Hoshangabad Road, Indrapuri).

PROVIDER	FLEET & BHOPAL REACH	DELIVERY SLA	PROOF OF DELIVERY	RATE MODEL	VERDICT
Shadowfax Hyperlocal	Established fleet operating across all major Bhopal zones	2 – 4 Hours (Same-day guaranteed)	Customer OTP	Contract-based (₹50 – 80 / drop)	Top Pick Powers Blinkit/Zepto
Pinified	Bhopal-native on-demand network with dedicated local riders	1 – 4 Hours (Instant dispatch)	App confirmation	Transparent rate (₹52 base up to 5km)	Strong Alternative Zero contract friction
Shiprocket Quick	Hyperlocal module bundling Dunzo, Shadowfax, Borzo	Same-Day (Rider dependent)	Courier dependent	Dynamic distance (₹10+ / km)	Convenient Variable rider density
Adloggs	AI-driven dispatch OS connecting local logistics providers	2 – 4 Hours	OTP / Photo	Quote based	Emerging Bhopal active

Grocery Strategy: Execute a corporate agreement with Shadowfax as primary carrier for Bhopal grocery orders. Simultaneously maintain Pinified as an active secondary provider for immediate rate-card fallback.

SECTION 06

Logistics Pricing & Operational Cost Analysis

A comprehensive breakdown of per-shipment commercial rate cards and monthly operational expenditure models:

Standard E-Commerce Rate Card (Shiprocket)		Hyperlocal Grocery Rate Card (Bhopal)	
Weight & Destination Zone	Indicative Cost	Delivery Distance (Store to Door)	Indicative Cost
500g • Local Bhopal intra-city	₹40 – 55	0 – 5 km (Core Neighborhood)	₹52 – 65
500g • Regional (MP & neighboring states)	₹60 – 80	5 – 10 km (Inter-zone Bhopal)	₹75 – 90
500g • Metro to Metro (Pan-India)	₹80 – 110	10 – 20 km (Outer Bhopal suburbs)	₹110 – 130
Additional 500g slab	₹35 – 50	Shadowfax B2B Base (Contract)	₹50 – 80 flat
Cash-on-Delivery (COD) surcharge	₹40 – 50 or 1.5%	3-Wheeler Heavy/Bulk Grocery (>15kg)	+25% to 35%
Return to Origin (RTO on failed delivery)	₹40 – 80	Peak Hour / Rain Surcharge	10% to 15%

Monthly Logistics Expenditure Projection (500 Orders / Month)

Order Classification	Monthly Volume	Avg. Cost / Order	Logistics Outlay	Platform SaaS Fee	Net Monthly Cost
Standard Goods (Shiprocket)	400 shipments	₹75.00	₹30,000	₹799 (Pro Plan)	₹30,799
Grocery Hyperlocal (Shadowfax)	100 deliveries	₹70.00	₹7,000	₹0 (Direct Contract)	₹7,000
Total Combined Operations	500 orders	₹74.00 (avg.)	₹37,000	₹799	₹37,799 / month

100% Cost Recovery Strategy: Standard Indian e-commerce practice charges consumers a nominal delivery fee (e.g., flat ₹49 on orders under ₹499, and free shipping on orders above ₹499). At an average collected shipping fee of ₹65 across orders, Intrust generates ₹32,500/month in shipping revenue, reducing net platform delivery expense to under ₹5,300/month.

SECTION 07

Integration Cost & Engineering Investment

Engineering work breakdown structure to develop, test, and productionize the automated logistics pipeline:

ENGINEERING WORKSTREAM	SCOPE OF WORK	EFFORT	COST ESTIMATE
Shiprocket API Client	Auth token caching, order creation, rate quotes, shipping label PDF fetch	3 – 4 Days	₹12,000 – 18,000
Webhook Handler & Sync	HMAC verification, status parsing, invocation of <code>update_order_delivery_v3()</code>	2 – 3 Days	₹8,000 – 12,000
Hyperlocal Grocery Adapter	Shadowfax/Pinified client, rider dispatch, address geo-tagging, OTP workflow	3 – 4 Days	₹12,000 – 18,000
Product Delivery Tagging	Schema migration + cart routing (standard pan-India vs. local grocery)	1 – 2 Days	₹4,000 – 8,000
Merchant Dashboard “Ship” UI	Replaces manual tracking number input with 1-click “Pack & Ship” button	2 – 3 Days	₹8,000 – 12,000
Customer Tracking Widget	Live milestone timeline showing courier status and external tracking link	2 Days	₹6,000 – 10,000
Idempotency & Error Guards	Duplicate shipment protection, webhook retry dead-letter queues	2 Days	₹6,000 – 10,000
Testing, Sandbox & QA	Sandbox courier verification, WhatsApp alert regression testing, edge cases	3 Days	₹10,000 – 15,000
Total Engineering Scope	Full 2-Provider Logistics Engine	18 – 21 Days	₹66,000 – 1,03,000

Phase 1: Standard E-Commerce Only

Automates pan-India delivery via Shiprocket (10–12 Days)

Automates 25+ couriers, AWB creation, label printing, and webhook status syncing for all non-grocery orders.

Dev Investment: ₹40,000 – 60,000

Phases 1 + 2: Complete Delivery Suite Full Scope

Automates both national shipping & Bhopal grocery delivery (18–21 Days)

Adds Shadowfax/Pinified hyperlocal routing, OTP verification, and multi-vendor delivery logic.

Dev Investment: ₹80,000 – 1,20,000

Annual System Maintenance: We advise budgeting approximately 20% of initial dev cost (~₹15,000 – 25,000 / year) to accommodate upstream carrier API schema updates, security token refreshes, and seasonal peak monitoring.

SECTION 08

Phased Rollout Roadmap (5 Weeks)

1

WEEK 1 • DAYS 1 – 5

Account Setup & Shiprocket Client Module

Register Shiprocket Advanced account, configure sandbox API tokens, build authentication and shipment generation modules.

2

WEEK 2 • DAYS 6 – 10

Webhook Synchronization & Merchant UI

Build webhook listener connected to `update_order_delivery_v3()` ; replace manual tracking inputs with 1-click “Pack & Ship” button in merchant admin.

3

WEEK 3 • DAYS 11 – 14

Customer Tracking UI & Phase 1 Production Go-Live

Launch live customer tracking page, verify automated WhatsApp notifications, and deploy Phase 1 to production for standard e-commerce orders.

4

WEEKS 4 – 5 • DAYS 15 – 21

Hyperlocal Grocery Integration & Dual-Routing

Deploy Shadowfax/Pinified hyperlocal adapter, configure delivery category tags on grocery items, test OTP delivery confirmation, and launch full suite.

SECTION 09

Risk Management & Mitigation Strategy

Return to Origin (RTO) on COD

Mitigation: WhatsApp OTP order confirmation before packing, mandatory landmark field at checkout, and automated Shiprocket NDR re-attempt workflows.

Medium Risk

Bhopal Peak Hour Rider Delays

Mitigation: Dual carrier setup: Shadowfax as primary contract carrier with Pinified configured as instant rate-card fallback during fleet crunches.

Medium Risk

Duplicate Shipments on Retries

Mitigation: Idempotency keys using internal `order_id` ; query existing AWB before invoking booking API on retry attempts.

Low Risk

Carrier API Schema Changes

Mitigation: Isolate external API payloads behind standardized internal contract adapters; reserve 20% annual maintenance budget.

Low Risk

SECTION 10

Final Recommendation & Immediate Next Steps

Strategic Summary: We recommend proceeding with the **two-phase execution plan**:

- Phase 1 (Weeks 1–3):** Automate pan-India standard shipping via **Shiprocket** (fastest ROI, eliminates manual tracking entry).
- Phase 2 (Weeks 4–5):** Activate **Shadowfax / Pinified** for Bhopal grocery items (sub-4-hour local delivery).

IMMEDIATE ACTION ITEM	STAKEHOLDER	REQUIREMENT	TIMELINE
1. Shiprocket Business Account Registration	Business Operations	Company GST & KYC Documentation	1 – 2 Days
2. Shadowfax Corporate Outreach	Business Operations	Bhopal Volume Projection (50–100 drops/mo)	2 – 3 Days
3. Engineering Sprint 1 Kickoff	Engineering Team	Client Proposal Approval	Upon Signoff